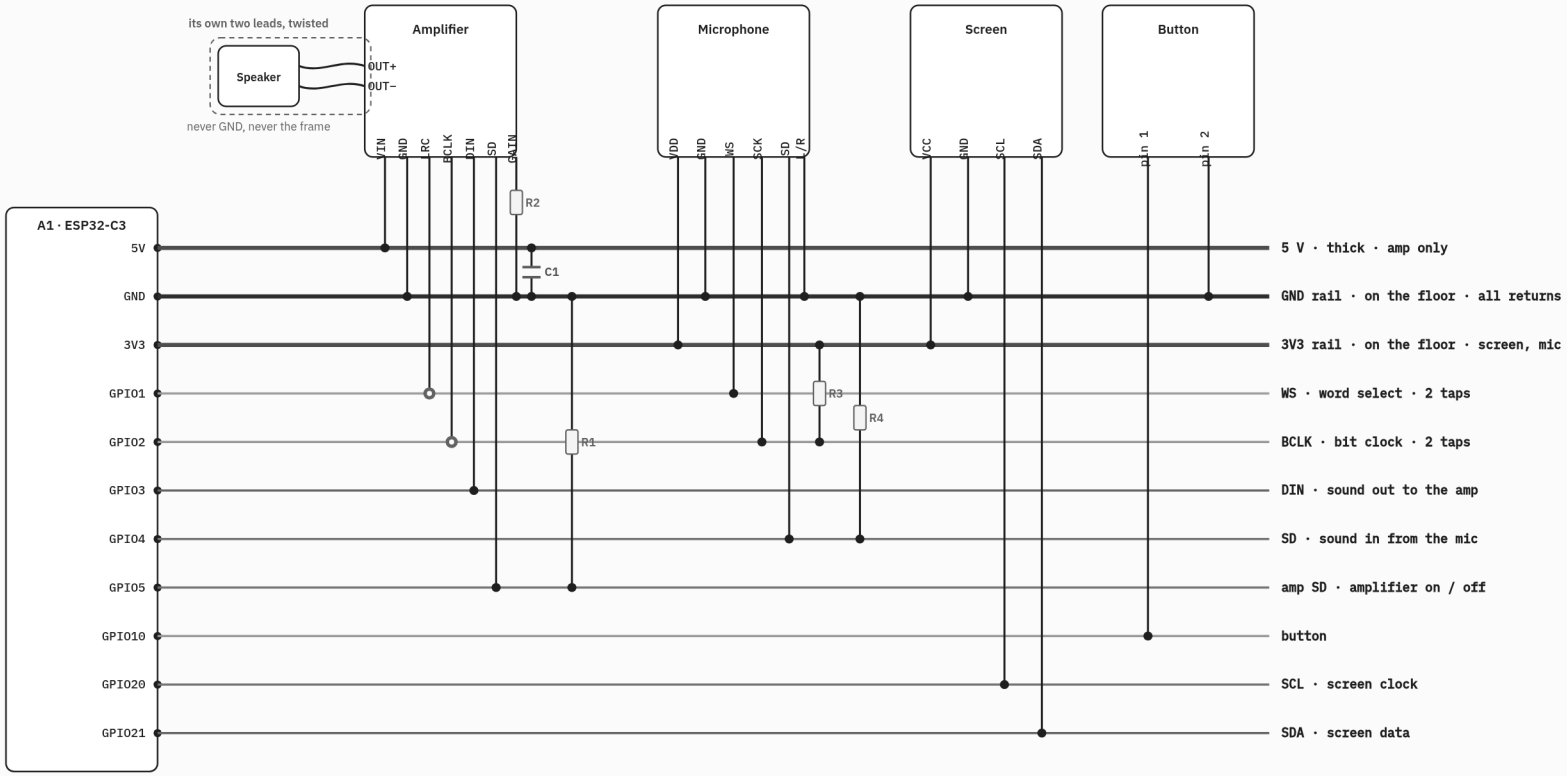
POCKET AI ASSISTANT · EVERY CONNECTION, FLAT

Pip wiring map

The whole device as one ladder. Each horizontal lane is one rod (or one of the two rails) leaving a pin on A1. Each part is a column above the lanes, touching only the lanes it needs. Scroll sideways if your screen is narrow; the drawing is wide on purpose.



Read any wire by following its lane. Start at the A1 pin on the left, run right, and every dot you cross is a part soldered to that rod. The two rails (GND, 3V3) are the rods lying on the floor of the frame; every other lane is one point-to-point rod.

The same thing as a list

FROM A1	LANE	ENDS AT	TAP ON THE WAY	PARTS ON THIS LANE
5V	5 V	amp VIN		220 µF + 100 nF across VIN and GND, right at the amp
GND	GND rail	amp GND, mic GND, mic L/R, screen GND, button contact 2		the return for everything; the frame is never part of it
3V3	3V3 rail	screen VCC, mic VDD		3.3 V only; never the amp
GPIO1	WS	mic WS	amp LRC	one rod, two taps
GPIO2	BCLK	mic SCK	amp BCLK	one rod, two taps; 10 kΩ from the mic's SCK to the 3V3 rail (boot strap)
GPIO3	DIN	amp DIN		Pip's voice going out
GPIO4	SD	mic SD		your voice coming in; 100 kΩ to the GND rail
GPIO5	amp SD	amp SD		4.7 kΩ pull-down to the GND rail keeps the amp off through reset
GPIO10	button	button contact 1		pressed = joined to the GND rail
GPIO20	SCL	screen SCL		
GPIO21	SDA	screen SDA		
amp GAIN		100 kΩ to the GND rail		sets 15 dB; no wire to A1
amp OUT+ / OUT−		speaker, its own two leads		never GND, never the frame

Thirteen connections, eleven rods leaving A1 (two of them with a tap), plus the two rails and the speaker's own leads. The lanes are drawn in a tidy order; in the frame the rails lie on the floor of the channel and the rods pass above them at stub height, see the Brass Wiring page for where each one physically goes.

Parts

PRODUCT YOU BOUGHT	HERE	LINK
Meshnology ESP32-C3 SuperMini (A1)	the left box	B0F888JQ91
HiLetgo MAX98357A amplifier	Amplifier column	B0CDWXZZCH
Same Sky CES-20134-088PM speaker	Speaker	DigiKey
AITRIP INMP441 microphone	Microphone column	B092HWW4RS
Hosyond SSD1306 OLED	Screen column	B09T6SJBV5
QTEATAK tactile button	Button column	B0FHW6HMG4
K&S #9861 1.0 mm brass rod	every lane	B005WPB7YG
LuminologyPro resistor kit	10 kΩ, 100 kΩ ×2, 4.7 kΩ	B0F4P352BB
ALLECIN electrolytic (220 µF) · BOJACK ceramic (100 nF)	at the amp	owned · no link recorded

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Sources. Connections from `plan/FAST_TRACK.md` Steps 6–11 and the hardware contract in `CLAUDE.md`; the 4.7 kΩ pull-down is the Step 11 default (1 kΩ if your board had a 100 kΩ pull-up and you chose that); the 100 kΩ GAIN value is the 15 dB setting from 2026-09-30. Same net list as the Wiring Plan and Brass Wiring pages, drawn flat.

What is and is not verified. This is the circuit you ran on the breadboard, redrawn; the brass version of it has not been built yet.